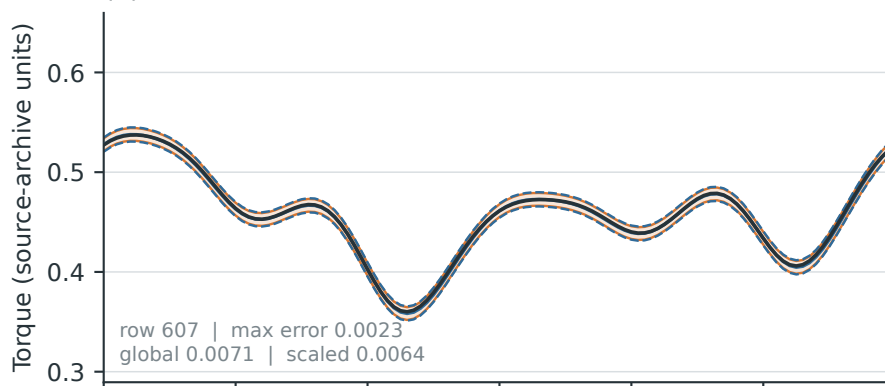


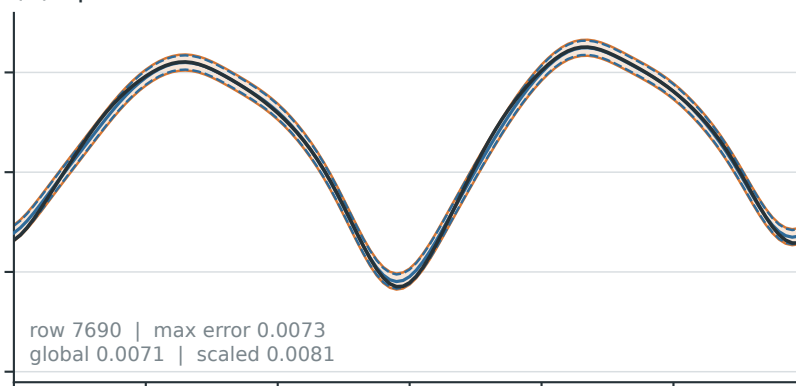
Representative 90% simultaneous torque bands

— High-fidelity torque — Fourier surrogate - - - Global boundaries — Geometry-scaled band

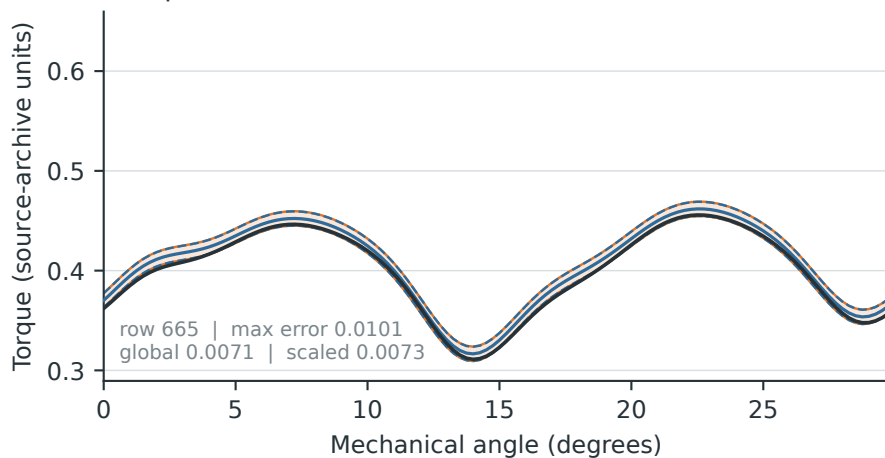
(a) Dense uniform: both cover



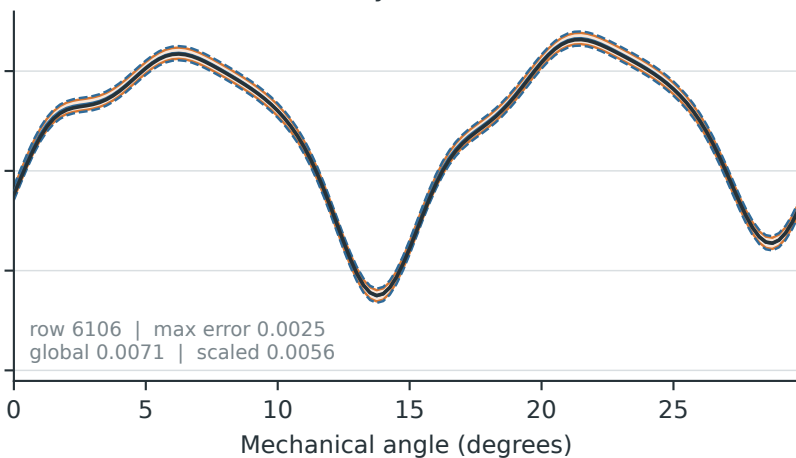
(b) Sparse uniform: scaled band rescues



(c) Sparse uniform: both miss



(d) Gaussian middle-density: both cover



Examples are selected by deterministic role and median-error rules; bands cover a design only when all 120 angles lie inside.